Ternary derivative controls and Reed-Muller weight caps: C44 < 2.1974348818

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Candidate, unrefereed. This result was found by AI agents working under my direction. I have only reviewed it at summary level, and other AI models reviewed it, so no human has checked it line by line.
The verifiers check the arithmetic, not the written lemmas.

Statement. *For the constant in* $r(f) \le C\, 2^{\deg(f)}$,

$$C_{44} \le \frac{1740987279236787915969864029 33}{7922816251426433759354395 0336} < 2.1974348818.$$

The lower bound is 1.5.

Editorial remark. The comparison with the exact rational is non-strict, as in the frozen proof's equation (12). The source README states it as strict; no strictness argument at the rational endpoint is supplied here. Since the rational is strictly below 2.1974348818, the decimal bound $C_{44} < 2.1974348818$ holds either way.

The seven numbered sections below transcribe the frozen `PROOF.md`. The exception is Section 4, which was expanded step by step and reworded for clarity on 6 November 2026, as marked there. Marked corrections reproduce the supplied errata without silently changing the original argument. Supplemental material follows Section 7.

Frozen source title: Ternary derivative controls and higher weight caps

Candidate, unrefereed. This is a written mathematical proof with exact arithmetic certificates, not proof-assistant formalization or independent human refereeing. Two separately written implementations check the finite artifact. Their finite controls do not prove the named external theorems. No novelty or priority is claimed. The framework is due to CHS/Wellens; the low binary weights are due to Berlekamp–Sloane and Kasami–Tokura.

1 Definitions, support, and extremizers

The cube has independent uniform sign coordinates. Every real function has a unique multilinear Fourier expansion $f = \sum_S \widehat{f}(S) \prod_{i \in S} x_i$. Its real degree is the largest $|S|$ with nonzero coefficient. A coordinate is relevant precisely when some such S contains it. Its local degree d_i is the largest $|S|$ containing it. Set

$$\begin{aligned} D(f) &= \sum_{i \text{ relevant}} 2^{-d_i}, \\ R_d &= \max \text{ number of relevant coordinates at degree at most } d, \\ W_d &= \max D(f) \text{ at degree at most } d. \end{aligned}$$

A nonzero multilinear real polynomial of degree at most k is nonzero on at least a 2^{-k} fraction of the cube. Here is an induction proof. The case $k = 0$ is constant. For a nonconstant polynomial choose a coordinate j and write $P = x_j A + B$ with A nonzero and degree at most $k - 1$. The induction hypothesis gives support density of A at least $2^{-(k-1)}$. For each input in that support, the two values $B + A$ and $B - A$ cannot both vanish. Averaging the two j branches gives support density at least 2^{-k}. This also handles polynomials whose degree is strictly below the stated upper bound.

For a sign Boolean f, write

$$A_i = \frac{f(i = +1) - f(i = -1)}{2}.$$

Its values are $0, +1, -1$, its real degree is $d_i - 1$, and $p_i = \mathbb{E} A_i^2 = \mathrm{Inf}_i(f)$. The support lemma gives $p_i \geq 2^{1-d_i}$. Parseval gives

$$I(f) = \sum_i p_i = \sum_S |S| \widehat{f}(S)^2 \leq \deg(f).$$

Consequently $D(f) \leq I(f)/2 \leq d/2$ and every relevant variable has influence at least 2^{1-d}. Thus $R_d \leq d\, 2^{d-1}$. After removing dummy variables and relabeling, only finitely many Boolean functions need be considered. The two displayed maxima therefore exist.

We use the CHS identity $W_d = R_d/2^d$, and include its proof because its extremal reduction matters here. Choose f maximizing D at degree at most d. Let L be its relevant coordinates of local degree below d, and Y the others. If L is nonempty, take a disjoint copy L' and a fresh sign selector z, and form

$$h = \frac{1 + z}{2} f(x_L, Y) + \frac{1 - z}{2} f(x_{L'}, Y).$$

This is Boolean. All degree-d terms of the two f copies involve only Y, so those terms cancel in the coefficient of z. Hence $\deg(h) \leq d$. Every Y coordinate retains local degree d. Each original L coordinate and its copy have local degree exactly one more than before: a largest Fourier term containing that coordinate gives a noncanceling term also containing z. Their two weights sum to the original weight. The selector is relevant because the two copies differ for some inputs whenever L is nonempty. It adds positive weight. Thus $D(h) > D(f)$, a contradiction. An extremizer for D has every relevant local degree d, proving $W_d \leq R_d/2^d$. Conversely, an R_d extremizer has

$D \geq R_d/2^d$. This proves the identity.

We also need every coordinate of an R_d extremizer to have local degree d. In bit coordinates, replace a coordinate of smaller local degree by the AND of two fresh bits. The new polynomial has degree at most d, since terms using that coordinate increase degree by at most one, and other terms keep their degrees. Both fresh coordinates are relevant: fix the other fresh bit to 1 and use a witnessing pair of inputs for the original coordinate. All other relevant coordinates remain relevant, by realizing either value of the substituted AND. The relevant count increases by one, contradicting maximality. This proves the required all-local-degrees-d reduction for count extremizers separately.

The published exact result $R_3 = 10$ supplies $W_3 = 5/4$. The elementary influence bound supplies $W_1 \leq 1/2$ and $W_2 \leq 1$. The R_3 premise is the result for arbitrary degree-at-most-three Boolean functions, not a restricted coloring class; see Tarannikov–Kirienko, ePrint 2000/050, and Krotov–Valyuzhenich, `https://arxiv.org/html/2311.05566v2` , section 3. CHS's identity is Corollary 2.2 in the final Combinatorica paper, `https://par.nsf.gov/servlets/purl/10201042` ; the corrected duplication proof also appears in Chiarelli's thesis, Proposition 8.5 and Corollary 8.6, `https://sites.math.rutgers.edu/~zeilberg/Theses/JohnChiarelli.pdf` .

2 A general control bound for ternary polynomials

Let A take values $0, +1, -1$, have real degree at most $k \geq 1$, and have h relevant coordinates. Put $p = \mathbb{E}A^2$. For a relevant coordinate j,

$$\Delta_j A = \frac{A(j = +1) - A(j = -1)}{2}$$

is a nonzero real polynomial of degree at most $k - 1$. Its nonzero values belong to $\{+1/2, -1/2, +1, -1\}$, so their squared magnitude is at least $1/4$. By the support lemma,

$$\mathbb{E}(\Delta_j A)^2 \geq \frac{1}{4}\, 2^{-(k-1)} = 2^{-(k+1)}.$$

On the other hand Fourier orthogonality, first for each derivative and then summed over coordinates, gives

$$\sum_j \mathbb{E}(\Delta_j A)^2 = \sum_S |S| \widehat{A}(S)^2 \leq k \sum_S \widehat{A}(S)^2 = kp.$$

There are h nonzero summands, each at least $2^{-(k+1)}$. Therefore

$$h \leq \left\lfloor kp\, 2^{k+1} \right\rfloor. \tag{1}$$

For the derivative A_i of a degree-d Boolean function set $k = d - 1$, $M = 2^k$ and $\lambda = Mp_i$. Formula (1) becomes

$$h_i \leq \lfloor 2k\lambda \rfloor. \tag{2}$$

No affine change of coordinates over $\mathbb{F}_2$ is being used to infer real polynomial degree. Formula (1) is a direct real Fourier calculation. At $\lambda = 1$ it recovers the worst $2k$ budget in the minimum-support

classification of Valyuzhenich, Theorem 2(2), `https://arxiv.org/html/2303.10995` . That external classification is useful context but is not needed as a premise of (1).

3 Restrictions controlled by any nonzero derivative

For completeness, the weighted restriction inequality for a coordinate set H is

$$D(f) \le \sum_{j \in H} 2^{-d_j(f)} + \mathbb{E}_\alpha D(f_\alpha). \tag{3}$$

To prove it for one fixed coordinate j, consider another relevant coordinate i. If i survives in both branches, its local degrees cannot increase, so both branch weights are at least its original weight. If it survives in exactly one branch, its derivative factors as $(1 \pm x_j)/2$ times the nonzero branch derivative. Its original local degree is exactly one larger than its surviving branch local degree. Half that branch weight therefore equals the original weight. A relevant i cannot disappear in both branches. Sum these inequalities over i other than j. Applying them successively to coordinates in H, or equivalently inducting on $|H|$, proves (3) for the coordinates outside H; the original weights of H supply the first term. Disappearing coordinates contribute zero on their branches.

Correction: Read the preceding iteration as follows: iterate over H for each coordinate outside H, then charge H at its original weights. Charging each later coordinate at its weight after restriction gives an inequality in the wrong direction. This is correction 1 in `CORRECTIONS.md`, reproduced below.

Choose a count extremizer f with $N = R_d$ and all local degrees d. Choose one of its coordinates i and let H be the full relevant-coordinate set of A_i; write $h = |H|$. Decompose $f = x_i A_i(x_H) + B(x_H, x_Y)$. On any control input α with $A_i(\alpha) = \pm 1$, both x_i branches being Boolean forces $B = 0$ for every Y. Thus f_α is a signed dictator and has $D = 1/2$. The fraction of such control assignments is exactly p_i.

Every degree-d monomial meets H. For a monomial in B disjoint from H, its coefficient cannot change on restricting H, since every larger monomial containing it would have degree above d. But that coefficient must vanish on any active control assignment, since B vanishes identically there. This is a contradiction. A degree-d monomial involving i contains a nonempty mask of A_i (of size $d-1$), hence also meets H for $d \ge 2$. Consequently every f_α has degree at most $d-1$.

Let $c \ge W_{d-1}$ with $c \ge 1/2$. Formula (3), all original local degrees being d, and the dictator branches imply

$$\frac{N}{2^d} \le \frac{h}{2^d} + \frac{p_i}{2} + (1 - p_i)c.$$

Use (2) and $\lambda = Mp_i$ to obtain the essential inequality

$$\begin{aligned} N &\le F_d(c, \lambda) \\ &:= 2^d c + \lfloor 2(d-1)\lambda \rfloor + (1 - 2c)\lambda. \end{aligned} \tag{4}$$

Choose i of minimum influence. Since $\sum_i p_i \le d$,

$$1 \le \lambda \le \frac{dM}{N}. \tag{5}$$

Equations (4),(5) apply to the SAME chosen derivative. They do not assume that every low-weight derivative has the same normal form.

4 Exact energy grid and the higher low-weight spectrum

Editorial remark. Revised 6 November 2026. In the first version, this section opened with a short paragraph (frozen `PROOF.md`, lines 173–197). Steps 1–7 below write out every step of it. The conclusions are unchanged: equation (6), and the fact that the support of A_i is a Reed–Muller codeword of normalized weight λ. The paragraphs after Step 7 are reworded for clarity. They now treat the case $d = 2$, which the first version left implicit, and no mathematical claim is changed.

Section 3 bounds N in terms of the real parameter $\lambda \geq 1$ of one chosen derivative. This section shows that λ can take only special values. Equation (6) puts λ on the grid $(2/M)\mathbb{Z}$. The Reed–Muller weight theorem then limits the values below 2 to the finite list (8). Section 5 checks the values in (8) below 2 and the necessary grid representatives (9) at and above 2.

Throughout this section, f is a sign Boolean function of real degree at most d with $d \geq 2$, i is a relevant coordinate, $k = d - 1 \geq 1$, $M = 2^k = 2^{d-1}$ and $\lambda = Mp_i$, as in Section 3. We use two coordinate systems on the cube: sign coordinates $x_t \in \{+1, -1\}$ and bit coordinates $b_t \in \{0, 1\}$, related by

$$b_t = \frac{1 - x_t}{2}, \qquad x_t = 1 - 2b_t,$$

so $x_t = +1$ corresponds to $b_t = 0$. For a set T of coordinates write $x^T = \prod_{t \in T} x_t$ and $b^T = \prod_{t \in T} b_t$, with $x^\emptyset = b^\emptyset = 1$.

Step 1 (integer bit coefficients). Every real function u on $\{0, 1\}^n$ has a unique multilinear expansion $u(b) = \sum_T c_T\, b^T$. For a set U, let 1_U be the bit vector equal to 1 exactly on U. Then $b^T = 1$ at 1_U if $T \subseteq U$ and $b^T = 0$ otherwise, so

$$u(1_U) = \sum_{T \subseteq U} c_T \qquad \text{for every } U.$$

Möbius inversion on the subset lattice solves this triangular system:

$$c_T = \sum_{U \subseteq T} (-1)^{|T|-|U|}\, u(1_U).$$

(Substituting the first identity into the right side gives $\sum_{V \subseteq T} c_V \sum_{V \subseteq U \subseteq T} (-1)^{|T|-|U|}$, and the inner sum is 1 if $V = T$ and 0 otherwise.) Consequently, if u takes only integer values, every c_T is an integer. This applies to the bit-output function $g = (1 - f)/2$, whose values are 0 and 1, and to A_i, whose values are $0, +1, -1$.

Step 2 (the two degrees agree). Substituting $b_t = (1 - x_t)/2$ into b^T gives $\prod_{t \in T}(1 - x_t)/2$, a multilinear polynomial in the x_t of degree $|T|$. So a multilinear bit polynomial of degree at most e becomes a multilinear sign polynomial of degree at most e. Substituting $x_t = 1 - 2b_t$ shows the converse in the same way. Multilinear expansions are unique in both coordinate systems, so the bit degree and the real (Fourier) degree of any function on the cube are equal. In particular $g = (1 - f)/2$ has bit degree at most d, because its Fourier expansion is $\frac{1}{2} - \frac{1}{2}f$ and therefore has degree at most d.

Step 3 (the Fourier grid of f). Write $g = \sum_T c_T\, b^T$ with integer c_T (Step 1) and $c_T = 0$ whenever $|T| > d$ (Step 2). For $S \subseteq T$, the coefficient of x^S in $b^T = 2^{-|T|} \prod_{t \in T}(1 - x_t)$ is $(-1)^{|S|} 2^{-|T|}$. Since

$f = 1 - 2g$,

$$\widehat{f}(S) = [S = \emptyset] - \sum_{T \supseteq S} (-1)^{|S|}\, c_T\, 2^{1-|T|},$$

where $[S = \emptyset]$ is 1 for the empty set and 0 otherwise. Each nonzero term has $|T| \le d$, so $2^{1-|T|} = 2^{d-|T|} \cdot 2^{1-d}$ is an integer multiple of $2^{1-d} = 1/M$. The constant $1 = 2^{d-1} \cdot 2^{1-d}$ is also a multiple of $1/M$. Hence $M\widehat{f}(S)$ is an integer for every S.

Step 4 (the grid of A_i). Put the expansion $f = \sum_S \widehat{f}(S) x^S$ into $A_i = \big(f(i{=}+1) - f(i{=}-1)\big)/2$. Terms with $i \notin S$ cancel. A term with $i \in S$ becomes $\widehat{f}(S)\, x^{S\setminus\{i\}}$, since $(1 - (-1))/2 = 1$. So

$$\widehat{A_i}(S') = \widehat{f}\big(S' \cup \{i\}\big) \qquad (i \notin S'),$$

and $\widehat{A_i}(S') = 0$ when $i \in S'$. By Step 3, $z_{S'} = M\widehat{A_i}(S')$ is an integer for every S'.

Step 5 (λ in terms of the z_S). By Parseval,

$$\lambda = M\,\mathbb{E}A_i^2 = M \sum_S \widehat{A_i}(S)^2 = \frac{\sum_S z_S^2}{M}.$$

Step 6 (parity). At the all-$(+1)$ sign input every x^S equals 1, so $A_i(+1,\dots,+1) = \sum_S \widehat{A_i}(S) = \sum_S z_S/M$. The left side is 0, $+1$ or -1, so $\sum_S z_S = M\, A_i(+1,\dots,+1)$ is an integer multiple of M. Because $d \ge 2$, $M = 2^{d-1}$ is even, so $\sum_S z_S$ is even. For every integer z, $z^2 - z = z(z-1)$ is even, so $\sum_S z_S^2 \equiv \sum_S z_S \equiv 0 \pmod 2$. Write $\sum_S z_S^2 = 2m$ with m an integer. Step 5 gives $\lambda = 2m/M$, that is,

$$\lambda \in \frac{2}{M}\mathbb{Z}. \tag{6}$$

Step 7 (the support of A_i is a Reed–Muller codeword). Regard A_i as a function on n bit coordinates, adjoining unused coordinates if necessary so that $n \ge k + 2$. This leaves p_i, λ and the degree bounds unchanged, because each value is repeated equally often. Write A_i in bit coordinates, $A_i = \sum_T a_T\, b^T$. Its values are integers, so every a_T is an integer (Step 1). By Step 4, $\widehat{A_i}(S')$ is nonzero only when $\widehat{f}(S' \cup \{i\})$ is, so the real degree of A_i is $d_i - 1 \le k$. Hence $a_T = 0$ for $|T| > k$ (Step 2). Reduce each a_T modulo two. The result $\bar{A} = \sum_T (a_T \bmod 2)\, b^T$ is a polynomial over $\mathbb{F}_2$ of degree at most k. At any bit input, evaluating an integer polynomial and then reducing modulo two gives the same result as reducing the coefficients first and then evaluating. So $\bar{A}(b) \equiv A_i(b) \pmod 2$. Since $0 \equiv 0$ and $\pm 1 \equiv 1$, $\bar{A}(b) = 1$ exactly when $A_i(b) \ne 0$. Thus the indicator of the support of A_i is a word of the binary Reed–Muller code $\mathrm{RM}(k, n)$. It is nonzero because i is relevant. Its weight is $w = 2^n \Pr(A_i \ne 0) = 2^n\, \mathbb{E}A_i^2 = 2^n p_i$, using $A_i^2 = 1$ on the support. Relative to the minimum distance 2^{n-k} of $\mathrm{RM}(k, n)$, its normalized weight is $w/2^{n-k} = 2^k p_i = \lambda$. We reduce A_i itself. Using only the real-degree bound $\deg(A_i^2) \le 2k$ would give membership in $\mathrm{RM}(\min\{2k, n\}, n)$; reducing A_i directly gives the required order-k bound.

We use the Kasami–Tokura theorem on Reed–Muller codewords of weight less than twice the minimum distance (references below). For $k \ge 2$, any nonzero word of $\mathrm{RM}(k, n)$ with weight strictly below $2 \cdot 2^{n-k}$ has normalized weight

$$\lambda = 2 - 2^{-j}, \quad \text{for an integer } j \ge 0. \tag{7}$$

Since $w < 2 \cdot 2^{n-k}$ is the same as $\lambda < 2$, (7) applies exactly when $\lambda < 2$. The theorem also says which j occur for given n and k. We use only the fact that λ has the form (7), so we allow

every $j \geq 0$. As in Step 7, we have $n \geq k+2$. The theorem applies to words of algebraic degree at most k. A nonzero word of algebraic degree at most $k-1$ lies in $\mathrm{RM}(k-1,n)$, so its weight is at least 2^{n-k+1}, twice the minimum distance of $\mathrm{RM}(k,n)$. The theorem is due to Kasami–Tokura (1970), with Berlekamp–Sloane, DOI 10.1109/TIT.1970.1054545. A published explicit statement and the two normal forms appear in Carlet (2024), section 2, p.10613, `https://www.aimspress.com/aimspress-data/math/2024/5/PDF/math-09-05-518.pdf` .

For $k \geq 2$, combine (6) and (7). By (6), $M\lambda$ is an even integer, and by (7), $M\lambda = M(2-2^{-j}) = 2^{k+1} - 2^{k-j}$. For $j \leq k-1$ this is even. For $j \geq k$ it is either nonintegral ($j > k$) or odd ($j = k$). When $k = 1$ ($d = 2$), (6) says λ is a positive integer, so $\lambda < 2$ forces $\lambda = 1$, which is (8) below with $k = 1$. (The certificate in Section 5 starts at $d = 4$, so this case is recorded only for completeness.) Hence, for every $k \geq 1$, every possible value of λ below 2 lies in the finite set

$$\{2 - 2^{-j} : j = 0, \ldots, k-1\}. \tag{8}$$

The first terms are $1, 3/2, 7/4, 15/8, 31/16, \ldots$. For $\lambda \geq 2$ we use only the grid (6). No further information about Reed–Muller weights in $[2, 2.5)$ (Kasami–Tokura–Azumi) is used.[1]

5 Finite scalar optimization, with the floor treated correctly

Suppose c bounds W_{d-1}. To exclude $N \geq n$, (5) puts λ in $[1,t]$, where $t = dM/n$. Since the set of possible λ for $N \geq n$ is contained in this interval, it suffices to show every allowed λ in $[1,t]$ has $F_d(c,\lambda) < n$. This excludes all $N \geq n$ at once, not merely $N = n$.

Use $c \in [1/2, (d-1)/2]$, checked inductively in the certificate. Within a cell $h \leq 2k\lambda < h+1$, the value (4) is

$$2^d c + h + (1-2c)\lambda,$$

which is nonincreasing in λ. It is NOT safe to substitute t directly inside its floor. Below 2 we test every member of (8) not exceeding t. For $\lambda \geq 2$, the possible cell indices are $4k$ through $\lfloor 2kt \rfloor$. On the grid (6), the first value in cell h is

$$\lambda_h = \frac{2}{M}\left\lceil \frac{hM}{4k} \right\rceil. \tag{9}$$

Discard it if $\lambda_h > t$ or $2k\lambda_h \geq h+1$. Every other grid value in that cell has no larger budget. Thus (8),(9) form a complete finite cover of all possible cases. Every such budget is strictly below n in `caps.json`. The resulting $c_d = (n-1)/2^d$ is valid by (4),(5) and the CHS identity.

Starting with $c_1 = 1/2, c_2 = 1, c_3 = 5/4$, proceed through $d = 96$. The certificate uses each already established c_{d-1} before deriving c_d. There is no circular use of the eventual constant. Some initial ceilings are:

[1] The attribution to third and higher below-two weights is open. An AI referee's ablation finds that, given the grid (6), excluding normalized weights in $(1, 3/2)$ alone already gives the same rational, while the ablation numbers in Section 5 disagree. The bound does not depend on that attribution. See Correction 2, printed at the end of Section 5.

d	R_d ceiling	c_d
4	24	3/2
5	54	27/16
6	119	119/64
7	254	127/64
8	528	33/16
9	1083	1083/512
10	2199	2199/1024
11	4439	4439/2048
12	8928	279/128

The frozen packet stores the most dangerous finite case at each first excluded count, its complete case count, and a feasible scalar point at the preceding count. Checker B independently enumerates all floor cells; checker A uses an independently derived collapsed envelope. For the latter, write $a = 2c - 1 \in [0, k-1]$. Consecutive first grid values in occupied cells $h < h'$ differ by at most $(h' - h)/(2k) + 2/M$. Therefore their maximal budgets increase by at least

$$\begin{aligned}(h' - h)\left(1 - \frac{a}{2k}\right) - \frac{2a}{M} \\ \geq 1 - \frac{a}{2k} - \frac{2a}{M} > 0.\end{aligned}$$

The last strict inequality is checked exactly for every degree; universally it follows for $k \geq 3$ from $(k-1)/(2k) + 2(k-1)/2^k < 1$ (at $k = 3$ this is $5/6$; at $k \geq 4$ the first term is below $1/2$ and the second is at most $3/8$). Thus the rightmost occupied high cell suffices for checker A. It locates that cell from the last grid point below t and tests its first grid point. Checker B uses none of this envelope shortcut.

Two additional certified variants quantify the inputs. Dropping the RM gaps altogether and allowing a continuum $\lambda \geq 1$ gives a bound below 2.2952887883. Retaining only $\lambda = 1$ or $\lambda \geq 3/2$ gives a bound below 2.2108998048. The third and higher below-two weights give the displayed 2.1974348818 bound. Each variant has its own inductive caps and both checkers verify them. These are candidate, unrefereed consequences of the same written lemmas, not claims about the exact Boolean maxima.

Correction: The attribution to third and higher below-two weights is open. The referee's ablation finds that, given the energy grid, the single fact that no normalized weight lies in $(1, 3/2)$ already gives the same rational. Section 5's ablation numbers disagree, and this has not been resolved; the bound does not depend on that attribution. This is correction 2 in `CORRECTIONS.md`.

6 An elementary infinite tail

We need a global bound, not just a horizon calculation. The CHS/Shinkar–Tal lower family starts with $\Xi_1 = x$ and, on disjoint variables, uses

$$\Xi_d = \frac{s+t}{2}\Xi_{d-1}(x) + \frac{s-t}{2}\Xi_{d-1}(y).$$

On a sign input exactly one summand is active, proving Booleanity by induction. Its nonzero

Fourier masks have size d. The two input copies and the two fresh selectors remain relevant, so its count L_d satisfies $L_1 = 1$ and $L_d = 2L_{d-1} + 2$. Solving this recursion gives

$$L_d = 3 \cdot 2^{d-1} - 2 = 3M - 2.$$

Thus $R_d \geq 3M - 2 \geq 2M$ for $d \geq 2$. For the least-influence coordinate of its count extremizer, (5) now yields $\lambda \leq d/2$. Apply (4) with the ACTUAL $c = W_{d-1}$, discard its floor downward, and divide by 2^d. Since $1/2 \leq W_{d-1} \leq (d-1)/2$, the multiplier $2d - 1 - 2W_{d-1}$ is positive and at most $2d - 2$. Hence

$$\begin{aligned} W_d &\leq W_{d-1} + \frac{\lambda(2d - 1 - 2W_{d-1})}{2^d} \\ &\leq W_{d-1} + \frac{d(d-1)}{2^d}. \end{aligned} \tag{10}$$

No block-sensitivity theorem or inherited numerical block certificate is required for (10). For $H \geq 1$, write $d = H + j$ and sum $j \geq 1$. The elementary geometric-series identities are

$$\sum 2^{-j} = 1, \qquad \sum j 2^{-j} = 2, \qquad \sum j^2 2^{-j} = 6.$$

The first follows by summing the geometric series; applying $x\, d/dx$ to $\sum_{j \geq 1} x^j = x/(1-x)$ once and twice, then setting $x = 1/2$, gives the other two. Expansion of $(H + j)(H + j - 1)$ gives

$$\sum_{d > H} \frac{d(d-1)}{2^d} = \frac{H^2 + 3H + 4}{2^H}. \tag{11}$$

The computed caps are nondecreasing through H. Every smaller degree is therefore bounded by c_H; for larger degrees telescope (10) and apply (11). Consequently $C_{44} = \sup_d R_d/2^d$ satisfies

$$\begin{aligned} C_{44} &\leq c_{96} + \frac{9508}{2^{96}} \\ &= \frac{1740987279236787915969864029 33}{7922816251426433759354395 0336} \\ &< 2.1974348818. \end{aligned} \tag{12}$$

The exact improvement over round five's frozen rational is

$$\frac{7685285257003586193491606937 7}{7922816251426433759354395 0336} > \frac{1}{1000}.$$

This is an exact rational comparison. Rounding displayed decimals never enters the proof.

7 What the recurrence and the scalar limit do and do not establish

The inherited valid block recurrence is

$$
\begin{aligned}
&U(0,d) = 0,\\
&U(b,d) = \min\left(c_d, \frac{d}{2^d} + \max_{1\le j\le d} U(b-1,j)\right),\\
&U(b,d) = U(B_d,d) \quad \text{for } b > B_d.
\end{aligned}
$$

Its proof and raw block duals are retained in `baseline/PROOF.md`, sections 5–6. In outline, restricting a maximum monomial of size d costs $d/2^d$ in (3) and lowers block sensitivity by one: the nonzero full monomial coefficient guarantees nonconstancy on its coordinates at every outside input; this gives one sensitive block disjoint from all sensitive blocks of the restricted function. Induct on b and take the maximum over every degree $j \le d$. Freezing a column above B_d is valid because that class of functions cannot grow further.

Checker B rechecks every retained Épi moment certificate and all 37 Fourier-lattice duals before re-running this recurrence with the inherited B_d. Checker A separately computes integer recurrence rows using only $B_d = d^2$, a consequence of the external Wellens theorem $3\,\mathrm{bs}(\mathrm{bs}-1) \le 2(d^4 - d^2)$, Theorem 4.1, equation (4.3), `https://arxiv.org/html/2005.00566` . Both reach exactly c_d in EVERY final column through degree 96. Thus the current block inputs are inactive at the saturation rows of this run. The accepted global bound (12) has the stronger elementary tail (11) and does not rely on those block premises. The saturation thresholds and slack are in `analysis.json`. A smaller B_d can help only if it crosses below its saturation threshold, with other inputs fixed; a decrement that leaves it at or above that threshold has zero effect in this recurrence.

There is also a precise limit for the scalar relaxation just specified. Define its next count as the largest integer N for which (4),(5),(6),(8) are jointly feasible, allowing every high grid value ≥ 2. The packet's feasible scalar points at its count ceilings and the strict exclusions at the next integers prove that the computed counts are exactly optimal for THIS relaxation. Those scalar points are not Boolean existence witnesses.

Let c be the previous scalar cap. Its denominator divides 2^{d-1}, so $N_0 = 2^d c$ is an integer. The case $\lambda = 1$ has budget $N_0 + 2d - 1 - 2c \ge N_0$ and satisfies $N_0\lambda \le dM$ whenever $c \le d/2$. It belongs to every allowed low spectrum. Hence the next cap is at least c. Every feasible scalar point at the largest count also has $N \ge 3M - 2$, so the same estimates as in (10) bound the scalar increment by $d(d-1)/2^d$. The maintained upper condition $c_d \le d/2$ continues after $d = 96$ by induction, since this increment is less than $1/2$ for $d \ge 6$. Therefore the infinite scalar sequence exists, is nondecreasing, and its limit lies in

$$
\left[\frac{174098727923678791596986393425}{2^{96}},\ \frac{174098727923678791596986402933}{2^{96}}\right]. \tag{13}
$$

This pins its limit to $2.197434881723198\ldots$ with interval width $9508/2^{96} < 1.3\cdot 10^{-25}$. It quantifies the end of this particular energy and weight-spectrum relaxation. It is not an optimality theorem for all uses of Reed–Muller structure. Full signed real-degree compatibility, sharper control sets for actual higher-weight derivatives, correlations between derivatives, or new block bounds crossing the saturation thresholds could improve it. The known lower bound is $3/2$ from the family above, leaving a candidate gap about 0.6974348817. The remaining gap cannot be removed by merely iterating this already saturated scalar relaxation farther.

Related energy prior art: Potapov, *On the number of relevant variables for discrete functions*, Cryptography and Communications 17 (2025), 989–998, `https://link.springer.com/article/10.1007/s12095-025-00797-4` , section 5, also uses Fourier energy and changed edges to

bound ternary relevance. Its root-of-unity output encoding can double the real degree of a ternary polynomial in our encoding. We credit the related method; no novelty is claimed for the elementary real-energy argument in section 2.

Corrections

Corrections to `PROOF.md`

`PROOF.md` is frozen so that its hash matches the receipts. A referee found two wording defects. Neither changes the mathematics.

1. Section 3, "Applying them successively to coordinates in H, or equivalently inducting on |H|". Read literally, this charges each later coordinate at its weight after restriction, and that inequality runs the wrong way. The correct reading: iterate over H for each coordinate outside H, then charge H at its original weights.

2. Sections 4 and 5 credit the third and higher below-two weights for the final bound. The referee's ablation finds that, given the energy grid, the single fact that no normalized weight lies in (1, 3/2) already gives the same rational. Section 5's ablation numbers disagree with that, and this hasn't been resolved. Treat the attribution as open. The bound doesn't depend on it.

External premises are R3 = 10 and the second Reed-Muller weight 3/2. The Kasami-Tokura-Azumi 1976 results aren't a premise. The claim in `KTA_AUDIT.md` about a table omission is unverified.

External premises and cited results

The external premises identified in `CORRECTIONS.md` are $R_3 = 10$ and the second Reed–Muller weight 3/2 (no normalized weight lies in $(1, 3/2)$). The degree-three result is cited through Tarannikov–Kirienko, `https://eprint.iacr.org/2000/050`, and Krotov–Valyuzhenich, `https://arxiv.org/html/2311.05566v2`, Section 3. These results are cited, not reproved.

Section 4 cites the fuller below-two Kasami–Tokura spectrum, crediting Berlekamp–Sloane for low binary weights: Kasami–Tokura (1970), `https://doi.org/10.1109/TIT.1970.1054545`; see also Carlet (2024), Section 2, p. 10613, linked there. Correction 2 leaves the need for the higher below-two weights for this bound open.

Section 1 proves and cites the Chiarelli–Hatami–Saks identity. Section 7 cites Wellens's recurrence and block theorem, whose premises are not needed for (12). Valyuzhenich's minimum-support classification is context, rather than a premise of Section 2's energy inequality.

KTA audit caveat. The Kasami–Tokura–Azumi (1976) results are not a premise. `KTA_AUDIT.md` reports that its original PDF endpoint and the Elsevier API denied access, and that the full 1974 preprint was not reconstructed. Its claim about an omission in the final table is *unverified*, as `CORRECTIONS.md` explicitly states. The proof allows all energy-grid values at and above 2 and uses no KTA exclusions in the range 2 to 2.5. The audit's claimed higher-weight numerical gain is also subject to correction 2.

Certificates and reproduction

Pinned source folder:

`https://github.com/rpinho/math-certificates/tree/09c84234137d3fadee1620dbc1c2ca58d2756ce2/c44/reed-muller-caps`

The frozen `PROOF.md` SHA-256, checked against `MANIFEST.json`, is

`af599ddff4732631a1c0565d5a5b3a90510d53955436b085b55883d85dcc4c47`

The packet includes the following files and subfolders.

- Written sources: `PROOF.md`, `CORRECTIONS.md`, `README.md`, `PRIOR_ART.md`, `KTA_AUDIT.md`, and `review/`.
- Exact inputs and generation: `caps.json`, `candidate.json`, `generate.py`, `analyze.py`, `analysis.json`, and `analysis.log`.
- Checkers: `verify_caps_a.py` and `verify_caps_b.py`. Checker B also reads `baseline/epi_certificate.json` and `baseline/spectral_certificate_round3.json`; `baseline/` retains the inherited proofs and certificates.
- Receipts and hashes: `check_a.log`, `check_b.log`, `verification_a.json`, `verification_b.json`, `FINAL_RECEIPT.json`, `MANIFEST.json`, `SHA256SUMS`, and `PROCESS_LEDGER.json`.
- Finite controls: `control_boolean.cpp`, `control_ternary.cpp`, `control_corruption.py`, `control_corruption.json`, `boolean.log`, `ternary.log`, and `corruption.log`. `jobs/` retains the original reproduction wrappers.

From a writable copy of the pinned folder, using Python 3.10 or later (standard library), the two checker commands are

```
nice -n 10 python3 -B verify_caps_a.py candidate.json
nice -n 10 python3 -B verify_caps_b.py candidate.json
```

The scripts write `verification_a.json` and `verification_b.json` beside themselves. Both were replayed for this PDF and returned PASS with the rational in (12). They check arithmetic conditional on the written lemmas and external premises, not those universal statements themselves.

Credit and later work

Following the source README and prior-art note, credit goes to Chiarelli, Hatami and Saks for the weighted framework (including the corrected duplication in Chiarelli's thesis); Wellens for the recurrence; Berlekamp and Sloane, Kasami, Tokura and Azumi for the weight spectra; Valyuzhenich for minimum-influence structure; BakeLab/Épi for inherited exact polynomial certificates; and Shinkar and Tal for the lower family also credited in Section 6. The degree-three citations are due to Tarannikov and Kirienko, and Krotov and Valyuzhenich. Potapov's related ternary spectral-energy method is credited in Section 7. The source prior-art note and KTA audit also discuss Carlet, Rolland, Klein and Storme, Krotov's public discussion, Lou and Wang, and Wei et al.; these contextual references do not supply additional premises for (12). No novelty, priority, or optimality is claimed.

Later work in the same repository (`small-degree-classification`) gives $C_{44} < 1.9916994577$, also unrefereed; this PDF concerns the requested 2.1974348818 proof.